123.cpp | arrayop.cpp | hashingop.cpp | 214.cpp | 2674.cpp | pointers.cpp

hashingop.cpp > main()

```cpp
#include <iostream>
#include <vector>

using namespace std;

class ParkingLot {
private:
    int totalSlots;
    vector<int> slots;

public:
    ParkingLot(int size) {
        totalSlots = size;
        slots.assign(totalSlots, -1);
    }

    int hashFunction(int carNumber) {
        return carNumber % totalSlots;
    }

    void parkCar(int carNumber) {
        int index = hashFunction(carNumber);
        int startIndex = index;

        while (slots[index] != -1) {
            if (slots[index] == carNumber) {
                cout << "Car " << carNumber << " is already parked.\n";
                return;
            }

            index = (index + 1) % totalSlots;

            if (index == startIndex) {
                cout << "Parking lot is full! Cannot park car " << carNumber << ".\n";
                return;
            }
        }

        slots[index] = carNumber;
        cout << "Car " << carNumber << " successfully parked at slot " << index << ".\n";
    }
```

```
        }

        slots[index] = carNumber;
        cout << "Car " << carNumber << " successfully parked at slot " << index << ".\n";
    }

    void displayLot() {
        cout << "\n--- Current Parking Lot Status ---\n";
        for (int i = 0; i < totalSlots; i++) {
            cout << "Slot " << i << ": ";
            if (slots[i] == -1) cout << "Empty\n";
            else cout << "Car " << slots[i] << "\n";
        }
    }
};

int main() {
    ParkingLot lot(5);

    lot.parkCar(102);
    lot.parkCar(205);
    lot.parkCar(302);
    lot.parkCar(402);

    lot.displayLot();

    return 0;
}
```

Problems Output Terminal Debug Console Ports

```
PS C:\Users\Admin\Untitled Folder 1> cd "c:\Users\Admin\Untitled Folder 1\" ; if ($?) { g++ hashingop.cpp -o hashingop } ; if ($?) { .\hashingop }
Car 102 successfully parked at slot 2.
Car 205 successfully parked at slot 0.
Car 302 successfully parked at slot 3.
Car 402 successfully parked at slot 4.

--- Current Parking Lot Status ---
Slot 0: Car 205
Slot 1: Empty
Slot 2: Car 102
Slot 3: Car 302
Slot 4: Car 402
PS C:\Users\Admin\Untitled Folder 1>
```